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03/30/26

Inverse Methods: HW5 (EAS 6134 Problem Set 5)

Linear Inverse Problem: Data from Constable et al. (2018)

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1) a. The overdetermined Least-Squares problem involves constructing a coarse "mesh" of points along the northing distance (i.e., along the track of the AUV). To do so, we consider steps of 40 meters, yielding 21 data points upon which we wish to determine the solution. For comparison, the data itself contains 33 measurements, with an average step of 24.5 meters.

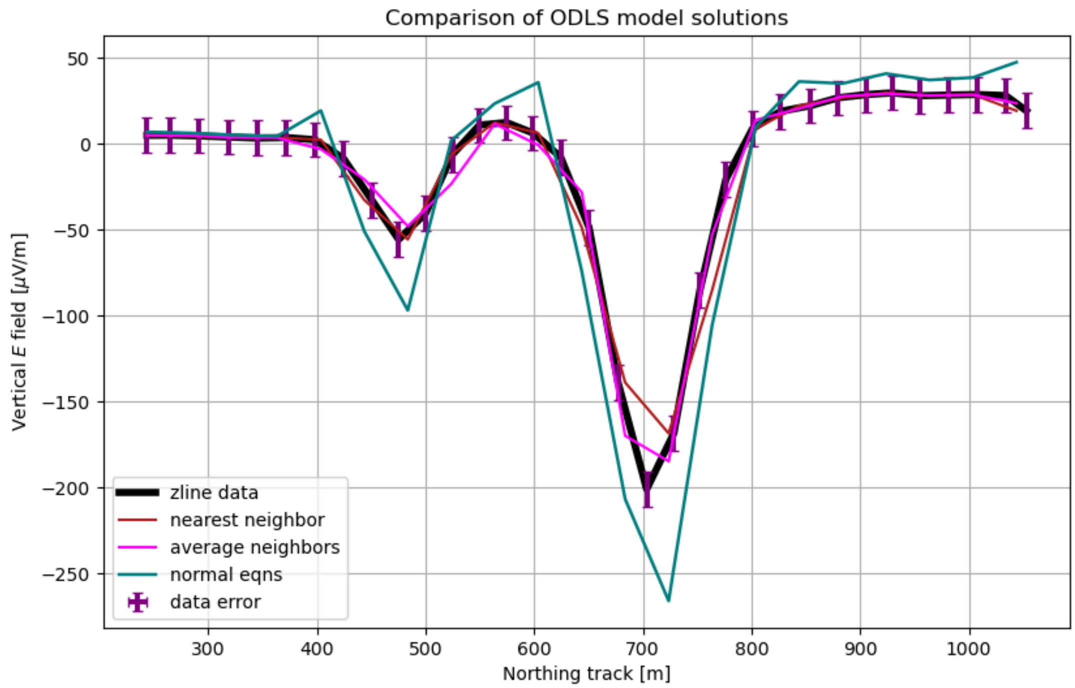
We employ the normal equations solution as a first step here. This involves calculating $\underline{\vec{m}}$ directly, via:

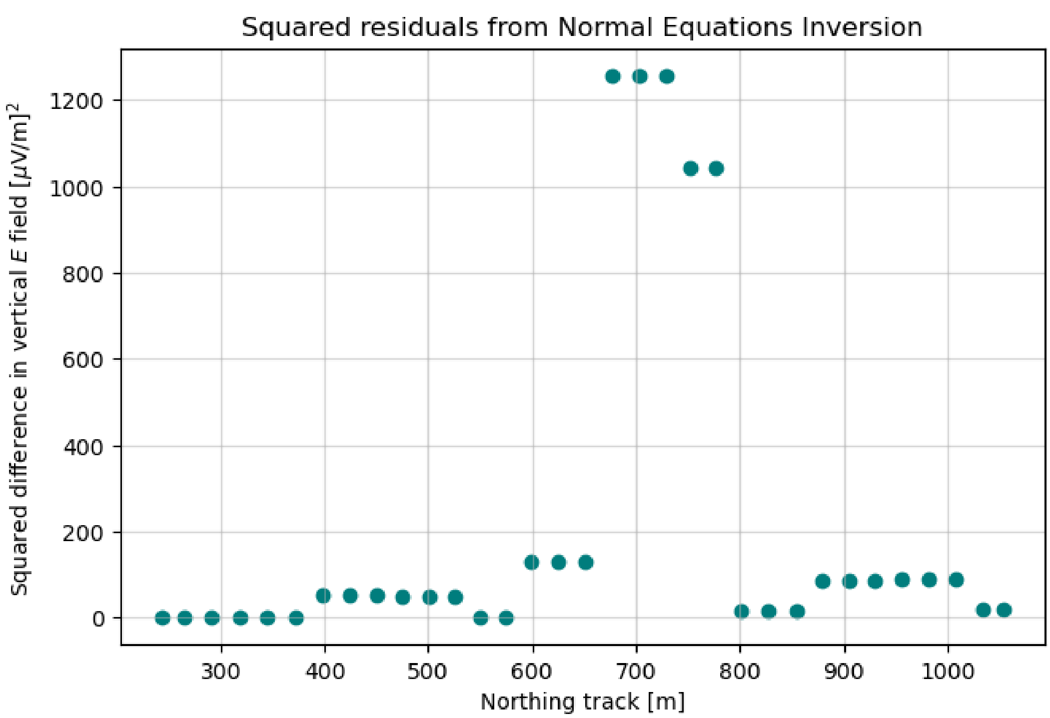
$$\vec{m}_{OOLS} = (\underline{\underline{G}}^T \cdot \underline{\underline{G}})^{-1} \cdot \underline{\underline{G}}^T \cdot \vec{d} \qquad (1)$$

where $\vec{d}$ is the observed data (i.e., the $\vec{E}$ field) and $\underline{\underline{G}}$ is a matrix that "averages" the two nearest data points around our (coarser) mesh. The solution to this inverse problem, via the normal equations (1), compared against the data as well as explicit extrapolation routines (nearest neighbor, linear interpolation between 2 nearest neighbors), are shown in the attached figure (following this page). The data, in black, include the noise associated with each measurement. The noise consists of stacking errors as well as a much larger (in magnitude) noise floor of 10 $\mu V/m$, employed by Constable et al. (2018) based on their knowledge of the instrumentation, measurement procedure, and any local anomalies. The stacking error is of order 0.1 $\mu V/m$, and is attributed with small variabilities in the actual measurements (i.e., not instrument-driven) that stem from normally-distributed, random, fluctuating signals.

The Normal equations solution, as seen in the figure (teal), captures the general behavior of the data but fails to fall within the noise floor compared to measurements in some locations. As indicated by the residuals, this solution is least effective near the strongest gradients in signal, and the residual plot demonstrates how the squared error grows quite large at these points.

1)a.


Comparison of ODLS model solutions
Vertical E field [μV/m]
Northing track [m]
zline data
nearest neighbor
average neighbors
normal eqns
data error
50
0
−50
−100
−150
−200
−250
300
400
500
600
700
800
900
1000


Squared residuals from Normal Equations Inversion
Squared difference in vertical E field [μV/m]^2
Northing track [m]
1200
1000
800
600
400
200
0
300
400
500
600
700
800
900
1000

1) b. To recast this problem as a linear inversion with more parameters than data, we construct a fine mesh, using a spacing of 10 meters, yielding 82 points upon which we wish to "interpolate" our data. We consider both a model norm minimizing solution, with the objective function

$$\vec{L}(\vec{m}) = \|\vec{m}\|^2 + \vec{\lambda}\cdot(\vec{d} - \underline{\underline{G}}\cdot\vec{m}), \quad (2)$$

as well as a model-derivative norm minimizing solution,

$$\vec{L}(\vec{m}) = (\underline{\underline{L}}\cdot\vec{m})(\vec{m}\cdot\underline{\underline{L}}) + \vec{\lambda}\cdot(\vec{d} - \underline{\underline{G}}\cdot\vec{m}), \quad (3)$$

where

$$\underline{\underline{L}} = \begin{pmatrix} -1 & 1 & 0 & \cdots & & 0 \\ 0 & -1 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & -1 & 1 & 0 \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 & -1 \;\; 1 \end{pmatrix} \Big/ \Delta x$$

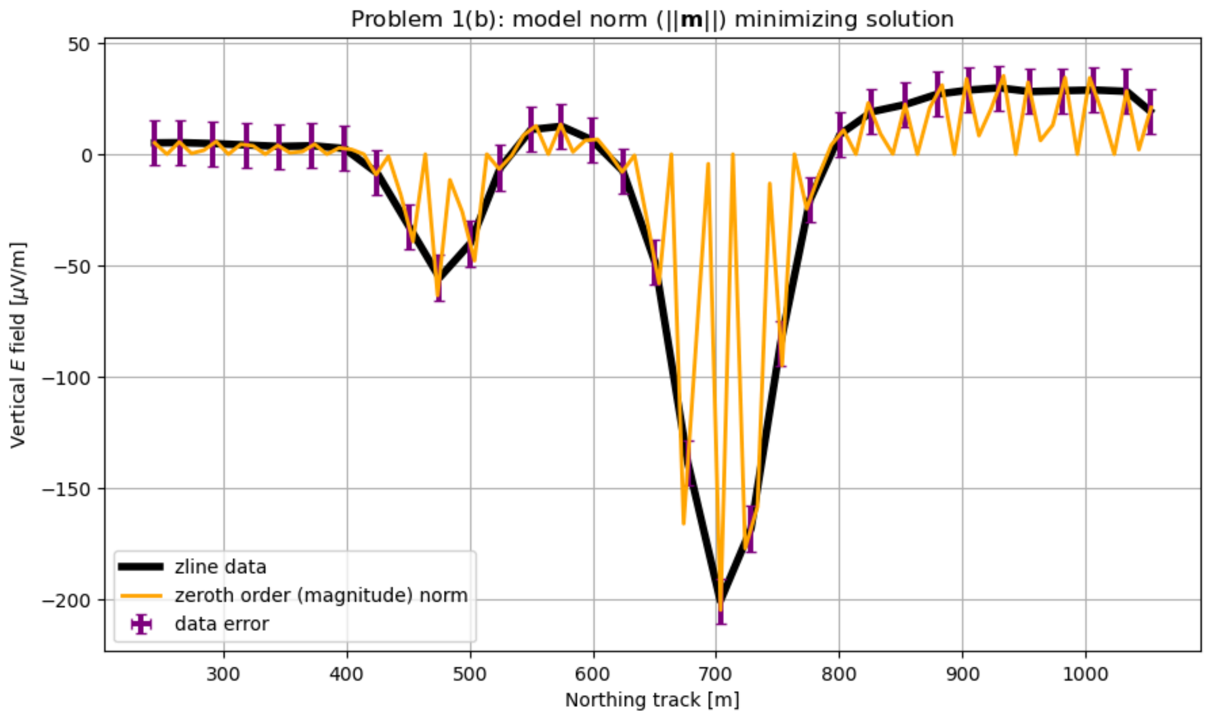
represents a finite difference operator to approximate the first derivative of the model. This will minimize the "roughness" of the model, rather than its magnitude. The analytical solutions to these inversions are included in a page following the plots. The two figures, representing each solution (orange: $\underline{\underline{L}}_0$, or $\|\vec{m}\|$ minimizing; teal: $\|\underline{\underline{L}}_1\cdot\vec{m}\|$ minimizing) are included below.

As indicated by the figures, the solution for the zeroth-order Tikhonov-regularized approach (2) is vastly over-fitted, i.e., since the model norm is a minimum, all values away from corresponding data points "spike" to nearly zero. However, as indicated by the residuals, the model does "fit" the data down to an accuracy of 10^{-29} $\mu V/m$, which certainly is below our computational limits even with double precision. However, the $\underline{\underline{L}}_0$ model clearly does not represent any realistic physical scenario.

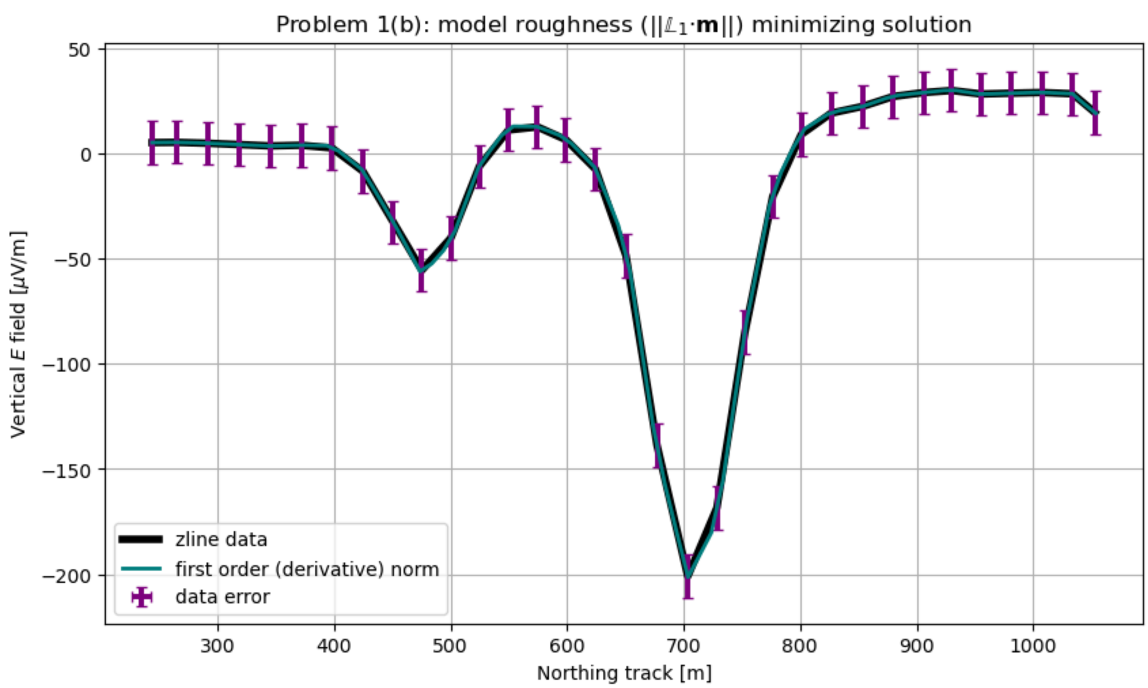
In contrast, the model-derivative norm minimizing solution, that is, Tikhonov first-order regularization, produces a far more realistic solution. This is because, as indicated by the signal, we have no reason to believe (physically-motivated) that the $\vec{E}$ field should be minimum magnitude outside of the observations. Rather, we would expect it to vary (on small scales) roughly quasi-linearly, which is enforced or targeted by including $\|\underline{\underline{L}}_1\cdot\vec{m}\|$ in our objective function. The residuals plotted below indicate the misfit is several orders of magnitude "worse" than the $\underline{\underline{L}}_0$ solution, but is also several orders of magnitude below even the stacking noise levels, and well below the noise floor. This solution is the more physically-motivated one.

Both solutions were computed numerically with numpy's linalg.inv() function. The model norm $\|\underline{\underline{L}}_1\cdot\vec{m}\| = 222.53$ $\mu V/m$ (for objective function (3)), compared to $\|\vec{m}_{OLS}\| = 392.37$ $\mu V/m$.

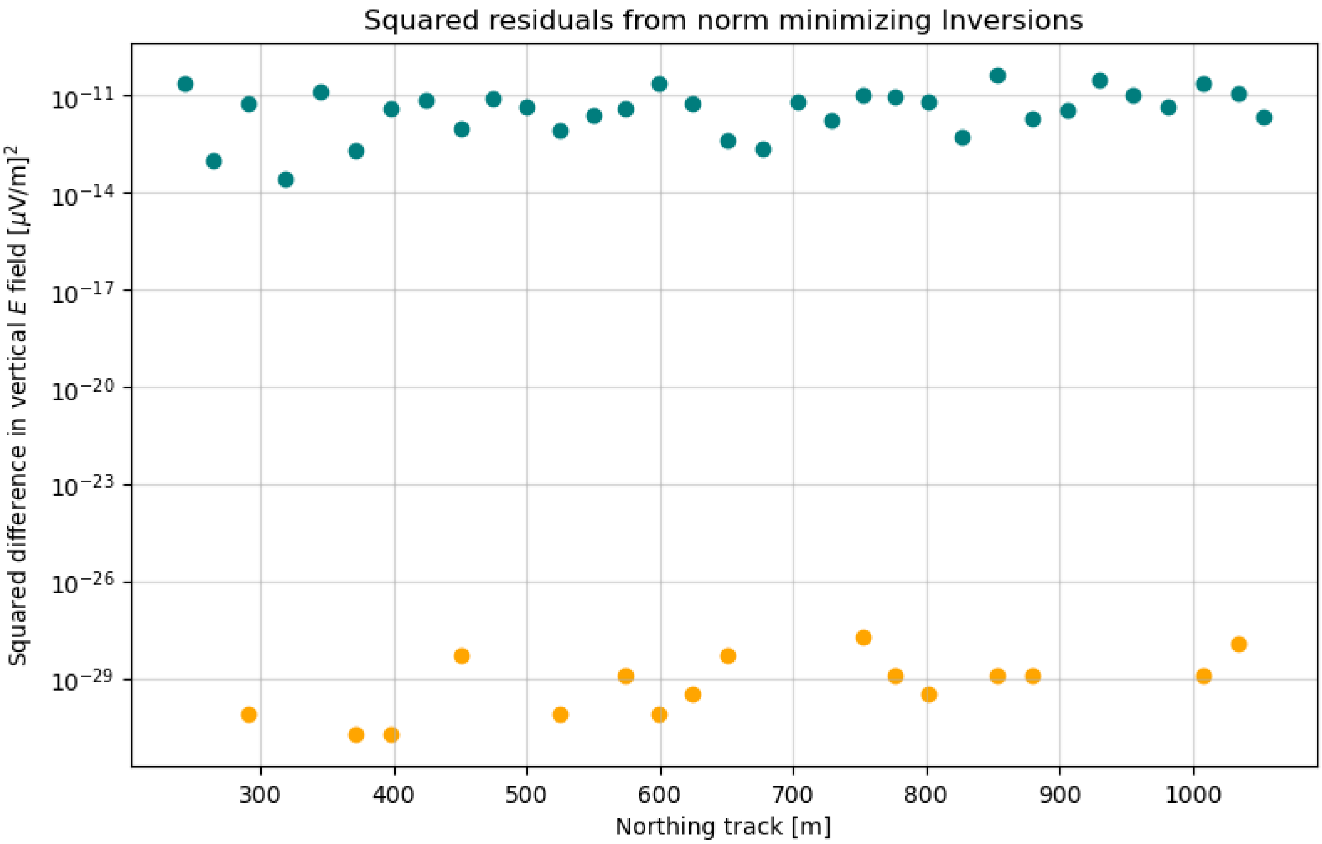
1)b.



Problem 1(b): model norm (||**m**||) minimizing solution
Vertical E field [μV/m]
Northing track [m]
zline data
zeroth order (magnitude) norm
data error
50
0
−50
−100
−150
−200
300
400
500
600
700
800
900
1000



Problem 1(b): model roughness (||$\mathbb{L}_1$·**m**||) minimizing solution
Vertical E field [μV/m]
Northing track [m]
zline data
first order (derivative) norm
data error
50
0
−50
−100
−150
−200
300
400
500
600
700
800
900
1000


Squared residuals from norm minimizing Inversions
Squared difference in vertical E field $[\mu V/m]^2$
10^{-11}
10^{-14}
10^{-17}
10^{-20}
10^{-23}
10^{-26}
10^{-29}
300
400
500
600
700
800
900
1000
Northing track [m]

Inverse Methods: Analytical Solutions

problem 1) b: underdetermined

$\Rightarrow$ need to include norm in objective function

$\rightarrow$ Start: Simply consider model norm:

$\vec{\lambda}$: vec of lagrange multipliers

$$L(\vec{m}, \vec{\lambda}) = \vec{m}^T \cdot \vec{m} + \vec{\lambda} \cdot (\vec{d} - \underline{\underline{G}} \cdot \vec{m})$$

$$\partial_{\vec{m}} L = \underline{2\vec{m} - \underline{\underline{G}}^T \cdot \vec{\lambda} = 0} \quad (1)$$

$$\Rightarrow \quad \vec{\lambda} = 2 (\underline{\underline{G}} \cdot \underline{\underline{G}}^T)^{-1} \cdot \vec{d}, \text{ since}$$

$(1) \Rightarrow \vec{m} = \frac{1}{2} \underline{\underline{G}}^T \cdot \vec{\lambda}$. substituting yields $\vec{\lambda}$, which then produces

$$\vec{m} = \frac{1}{2} \underline{\underline{G}}^T \cdot \vec{\lambda} = \frac{1}{2} \underline{\underline{G}}^T \cdot (2 \underline{\underline{G}} \cdot \underline{\underline{G}}^T)^{-1} \cdot \vec{d}$$

$$\Rightarrow \vec{m} = \underline{\underline{G}}^T \cdot (\underline{\underline{G}} \cdot \underline{\underline{G}}^T)^{-1} \cdot \vec{d}$$

$\rightarrow$ However, this is not physically motivated. Rather, we seek to minimize the "roughness", so we apply the <u>1st derivative model norm in our objective function:</u>

$$L(\vec{m}, \vec{\lambda}) = (\underline{\underline{L}} \cdot \vec{m}) \cdot (\underline{\underline{L}} \cdot \vec{m}) + \vec{\lambda} \cdot (\vec{d} - \underline{\underline{G}} \cdot \vec{m})$$

$$= \vec{m} \cdot (\underline{\underline{L}}^T \cdot \underline{\underline{L}}) \vec{m} + \vec{\lambda} \cdot (\vec{d} - \underline{\underline{G}} \cdot \vec{m})$$

again, differentiate:

$$\partial_m L = 2 (\underline{\underline{L}}^T \cdot \underline{\underline{L}}) \vec{m} - \underline{\underline{G}}^T \cdot \vec{\lambda} = 0$$

yielding

$$\Rightarrow \vec{m}(\vec{\lambda}) = \frac{1}{2} (\underline{\underline{L}}^T \cdot \underline{\underline{L}})^{-1} \cdot \underline{\underline{G}}^T \cdot \vec{\lambda}.$$

To determine $\vec{\lambda}$, substitute back in:

$$\vec{d} = \underline{\underline{G}} \cdot \vec{m} = \underline{\underline{G}} \cdot \left(\frac{1}{2} (\underline{\underline{L}}^T \cdot \underline{\underline{L}})^{-1} \cdot \underline{\underline{G}}^T \cdot \vec{\lambda}\right)$$

$$\Rightarrow \vec{\lambda} = 2 (\underline{\underline{G}} \cdot (\underline{\underline{L}}^T \cdot \underline{\underline{L}})^{-1} \cdot \underline{\underline{G}}^T)^{-1} \cdot \vec{d}$$

Substituting the expression for λ into our expression for $\vec{m}$ yields:

$$\vec{m}(\vec{\lambda}) = \frac{1}{2}\left(\underline{\underline{L}}^T \cdot \underline{\underline{L}}\right)^{-1} \cdot \underline{\underline{G}}^T \cdot \vec{\lambda} = \frac{1}{2}\left(\underline{\underline{L}}^T \cdot \underline{\underline{L}}\right)^{-1} \cdot \underline{\underline{G}}^T \cdot \left(2\left(\underline{\underline{G}}\left(\underline{\underline{L}}^T \cdot \underline{\underline{L}}\right)^{-1} \cdot \underline{\underline{G}}^T\right)^{-1} \cdot \vec{d}\right.$$

$$\Rightarrow \underline{\vec{m} = \left(\underline{\underline{L}}^T \cdot \underline{\underline{L}}\right)^{-1} \cdot \underline{\underline{G}}^T \cdot \left(\underline{\underline{G}} \cdot \left(\underline{\underline{L}}^T \cdot \underline{\underline{L}}\right)^{-1} \cdot \underline{\underline{G}}^T\right)^{-1} \cdot \vec{d}}$$

This solution minimizes the rate of change, or "roughness" of the model, as perscribed by the first-order differential operator $\underline{\underline{L}}$.

1) C. Now, we include a Lagrange multiplier (λ) in our objective function (different from the vector factor $\vec{\lambda}$ in (2,3)!) that "weights" the model norm and the residuals against each other in the magnitude of the objective function. We also employ a target misfit equal to $\varepsilon = 10\ \mu V/m$, since the stacking noise is 2-3 orders of magnitude weaker everywhere.

To determine the model which most accurately replicates the data (to within the target misfit) as well as contains a physically-motivated (i.e., linear) behavior between observations, we perform the inversion from (P1 b), equation (3), using many different Lagrange multipliers spanning from $\lambda \in [10^{-5}, 10^{5}]$.

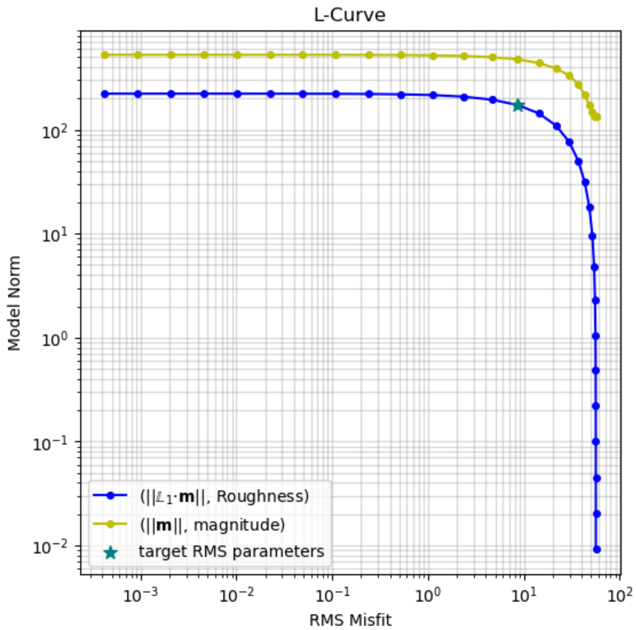
The Figures that follow depict the various properties of our fitting procedure for the target RMS misfit (see above) and with various values of λ. As indicated by the L-curve, the ~~does~~ combined with under determined nature of our procedure the $\|\underline{\underline{L}}_1 \cdot \vec{m}\|$ model norm does not yield the "nice" behavior of a global minimum distance to the origin at the target misfit. In other words, it is not an "L" shape. This is because the misfit remains high even as the $\|\underline{\underline{L}}_1 \cdot \vec{m}\|$ norm is minimized, because a flat line cannot match the observations. In turn, with a low norm, the same "overfitting" that occurred in 1(b) for the orange curve (objective function (2)) will be produced by the model, yielding a very high norm. The "kink" or "elbow" in the curve occurs where the target misfit is achieved, as indicated by the teal star. The other panel demonstrates that the RMS misfit follows the expected exponential (linear in log scale), then "flat" behavior with increasing λ value.

Next, we depict the minimum deviation from target misfit model in teal. As expected, the solution is similar to our model obtained in part (b), but it also includes slightly more relaxation or dampening to within the target misfit range ~ (error bars). Finally, we demonstrate all solutions obtained with our sweep over λ-space: Similar to the behavior indicated by the L-curve figure, the trade-off introduced by varying λ in our objective function produces "sharp" models that slightly over-fit or fit exactly (with low RMS misfit, which we know to be inaccurate) versus "flat" models that have minimal norms.

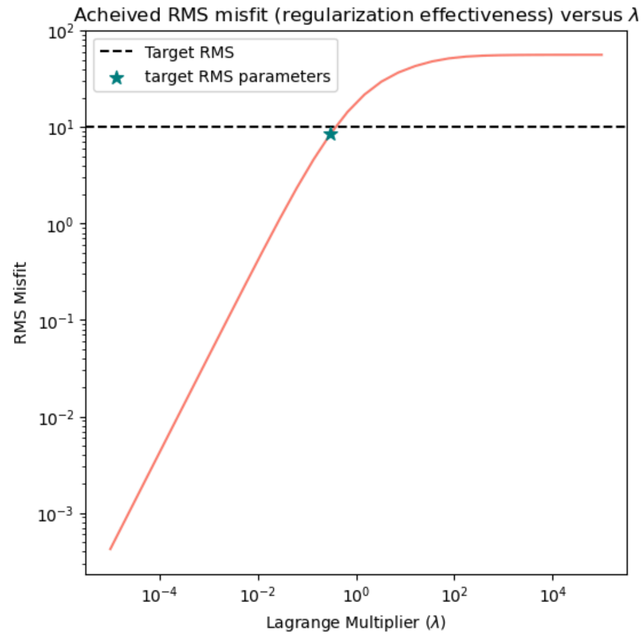
Tabulated values are included below:
(λ, RMS, $\|\underline{\underline{L}}_1 \cdot \vec{m}\|$, $\|\vec{m}\|$)

norm: L_1, i.e., $\|\underline{\underline{L}}_1 \cdot \vec{m}\|$

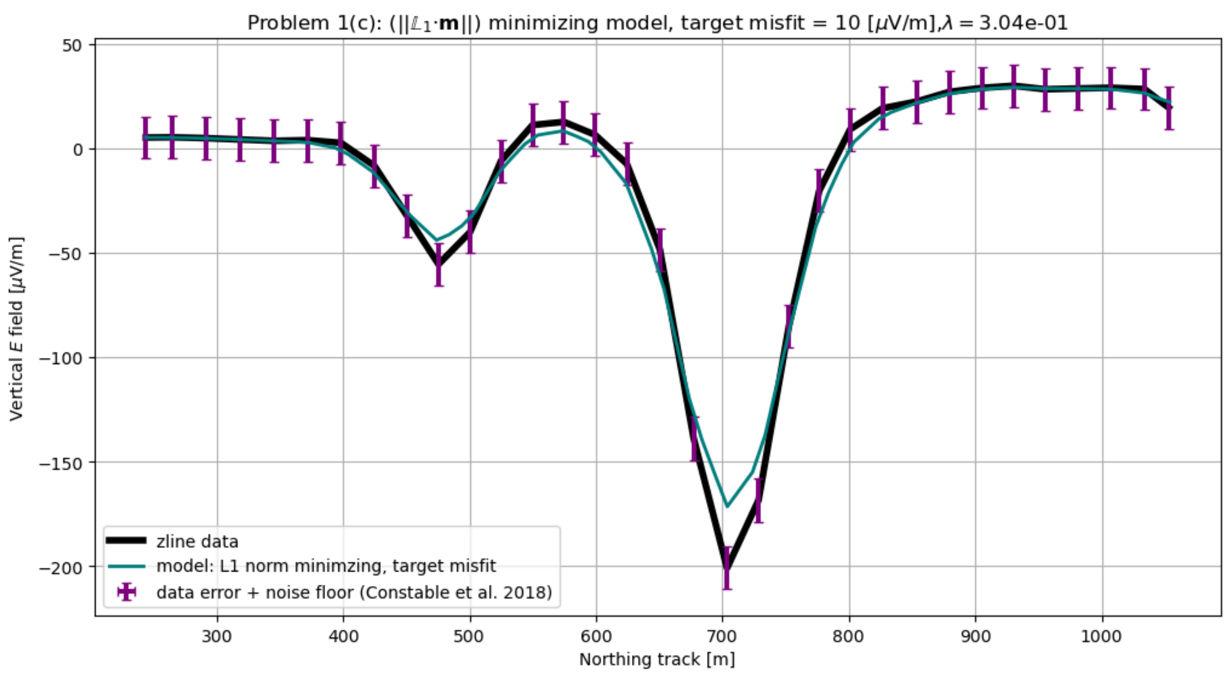
1)c.



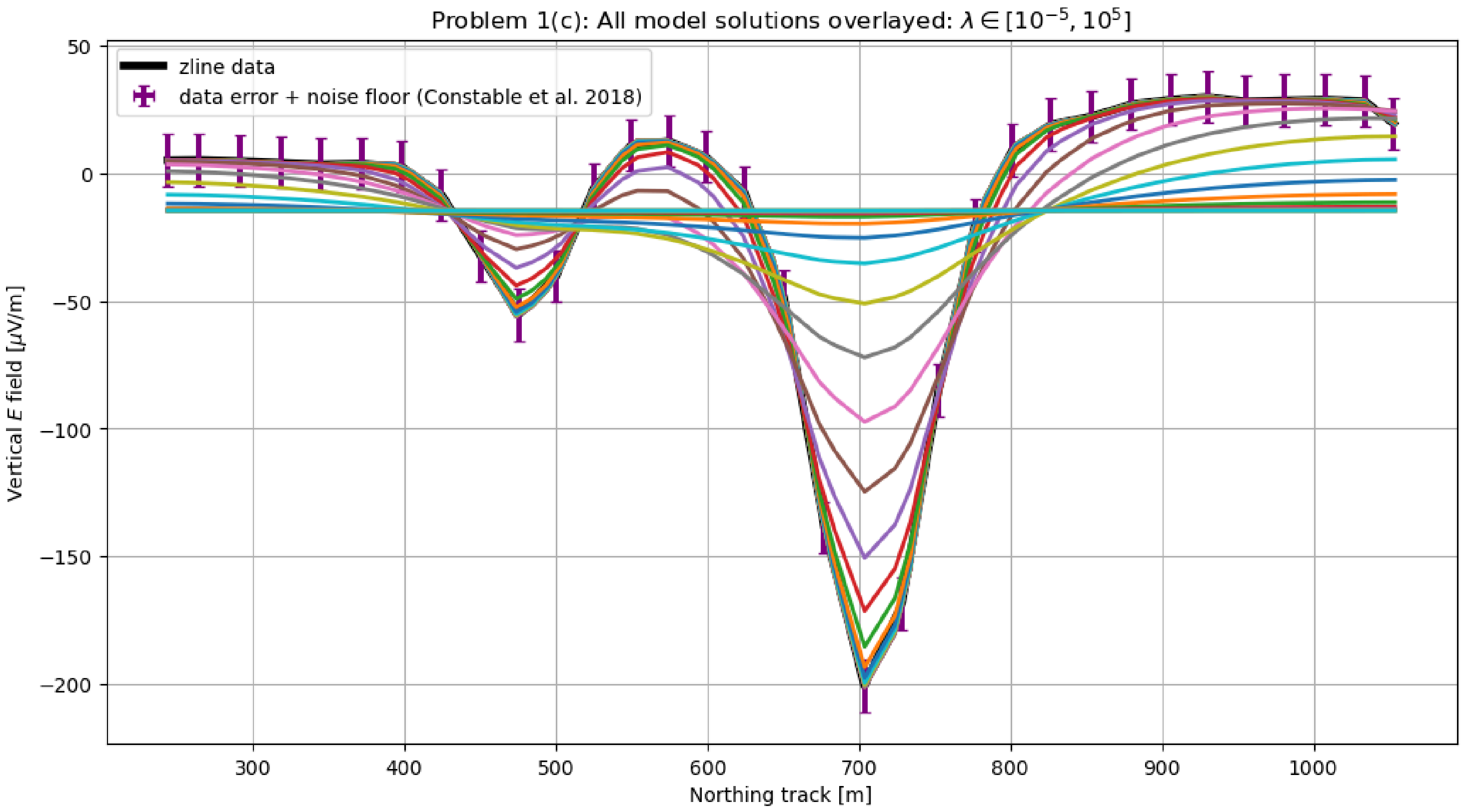
L-Curve
Model Norm
RMS Misfit
(||𝕃1·m||, Roughness)
(||m||, magnitude)
target RMS parameters



Acheived RMS misfit (regularization effectiveness) versus λ
RMS Misfit
Lagrange Multiplier (λ)
Target RMS
target RMS parameters



Problem 1(c): (||𝕃1·m||) minimizing model, target misfit = 10 [μV/m],λ = 3.04e-01
Vertical E field [μV/m]
Northing track [m]
zline data
model: L1 norm minimzing, target misfit
data error + noise floor (Constable et al. 2018)



Problem 1(c): All model solutions overlayed: $\lambda \in [10^{-5}, 10^{5}]$
zline data
data error + noise floor (Constable et al. 2018)
Vertical E field [μV/m]
Northing track [m]
50
0
−50
−100
−150
−200
300
400
500
600
700
800
900
1000

lambda	rms	model_norm	true_norm
0.00001	0.000421596107578405	222.53048203157	526.322087100765
2.21221629107045E-05	0.000932637074971495	222.527287007704	526.319281776773
4.8939009184775E-05	0.00206307405954588	222.520219662759	526.313076202758
0.000108263673387405	0.00456337472873602	222.504588781618	526.299350133697
0.000239502661998749	0.0100922804240976	222.470027562861	526.268994894246
0.000529831690628371	0.0223121818112851	222.393656960982	526.201890386742
0.00117210229753348	0.049290516305922	222.225128903795	526.053673962617
0.00259294379740467	0.108707214410717	221.854343030419	525.726919106996
0.00573615251044868	0.238879418907756	221.04381243894	525.009505266774
0.0126896100316792	0.520908674527513	219.296045198418	523.448023130002
0.0280721620394118	1.11833806440127	215.629858022329	520.109261681232
0.0621016941891562	2.3315371107244	208.325640484124	513.208110839286
0.137382379588326	4.62631034403521	194.969583754191	499.75699717016
0.30391953823132	8.53667972569045	173.39585906946	475.840889748437
0.672335753649934	14.361184502693	143.567548376675	438.51593242051
1.48735210729351	21.750454355701	109.104598765125	388.953357158899
3.29034456231267	29.6182422127253	76.361781148613	332.454972615732
7.27895384398316	36.8428825564754	50.2530111100495	273.647829947167
16.1026202756094	43.0088949464543	31.351336703477	216.944863398929
35.6224789026244	48.0338933816888	18.1662334296289	172.287359728479
78.8046281566992	51.6644492564679	9.66987708578389	147.2042116805
174.332882219999	53.8959435754977	4.80703243435913	137.199924156658
385.662042116347	55.1007700541753	2.28618029221683	134.046369226294
853.167852417281	55.6996508199956	1.0600514760737	133.153277393665
1887.3918221351	55.9835751668243	0.485038106979928	132.898595402797
4175.31893656041	56.1148782343673	0.220495635261214	132.818968382569
9236.70857187388	56.1748643794005	0.0999298439384314	132.790773372325
20433.5971785695	56.2021122456814	0.0452249640766888	132.779676276228
45203.5365636024	56.2144564824602	0.020454180866827	132.775001955635
100000	56.2200421044011	0.00924824355994123	132.772959361629

Problem 2: Generalizing the inversion to a map

To perform the 2-D inversion, we generalized our 1-D approach from problem 1(c):

→ We consider a 1st-derivative model norm minimization, or 1st-order Tikhonov regularizer, to obtain a physically appropriate "smoothness" of the $\vec{E}$-field data

→ We employ the scipy.kron (kronecker delta) function to generate the $\underline{L}_1$ arrays for each dimension

→ We use a locally-linear basis to construct $\underline{G}$, involving obtaining bi-linear weights, according to

$$V_1 = \frac{1}{x_F y_F}(V(x,y)), \quad V_2 = \frac{1}{(1-x_F)y_F}\cdot V(x,y), \quad V_3 = \frac{1}{(1-y_F)x_F}\cdot V(x,y),$$

$$V_4 = \frac{1}{(1-x_F)(1-y_F)}\cdot V(x,y),$$ for the schematic depicted in Fig 6 of Constable et al, 2018.

→ We store all matrices as scipy.sp.csr_matrix (i.e., sparse matrices) since the 100x100 discretization would otherwise cause large slowdown / memory utilization. We use a sparse matrix solving algorithm, scipy.sp solve (linear algebra module)

→ Again, the noise floor of 10 is much larger than the stacking errors, and dominates the uncertainty of the inversion. However, now the RMS misfit needs to be slightly smaller than the magnitude of the noise floor, since the problem is extremely sparse (few observations (626) compared to 10,000 points).

⇒ We use the L-curve analysis, following our logic from P1(c) (UDLS), to determine an appropriate choice of the noise floor. We select a target RMS misfit of $\frac{1}{2}\times$ noise floor, or 5 $\mu V/m$. This occurs where the RMS misfit begins to grow but is not substantially high (compared to the noise floor), but also where the model norm $\|\underline{L}^T\cdot\underline{L}\cdot\vec{m}\|$ begins to drop (i.e., it is far lower than its maximum value in λ-space).

(See attached figure)

→ Our solution produces most features of Constable et al., (2018), accounting for their sign change (see pg. 54 discussion). The features are sharper with a lower target misfit, but this is even farther below the noise floor from their work.